

Ibn Tofail University

Rigid body mechanics

Problem Set I

Exercise 1:

Let $(A, \vec{V})$ be a bound vector defined in the basis $(\vec{i}, \vec{j}, \vec{k})$ by: $A(-2, -1, 2)$ and $\vec{V} = \vec{i} + 2\vec{j} - \vec{k}$.

1. Calculate the moment of the vector $\vec{V}$ at the point with coordinates $(2, 1, 1)$.
2. Calculate the moment of $(A, \vec{V})$ with respect to the axis Δ passing through B and parallel to the vector $\vec{U} = 2\vec{i} - 3\vec{k}$.

Correction

Exercise 2:

We are given two torsors T_1 and T_2 . Find the set of points (if it exists) in space where their moments are collinear.

Correction

Exercise 3:

Let E be a three-dimensional vector space and S a linear application from E to E which associates to any vector $\vec{U}$ of E its image $\vec{U}' = S(\vec{U})$. We are interested in the case where S is an antisymmetric application defined by: $S(\vec{U}) = L \cdot \vec{U}$.

Find the matrix L associated with the antisymmetric application S by taking s_1, s_2 , and s_3 as components of the vector $\vec{S}$.

Correction

Exercise 4:

To every point $P(x, y, z)$ of affine space, we associate the family of vector fields $\vec{V}_t(P)$ defined by:

$$\vec{V}_t(P) = (3y - tz + 1, -3x + 2tz, tx + t^2y - 4/3); \quad t \in \mathbb{R}.$$

1. Show that the only equiprojective fields are obtained for $t = 0$ and $t = 2$.
2. Determine for each case the associated torsors by their reduction elements at point $O(0, 0, 0)$.

Correction

Exercise 5:

We are given two sliders $(A, \vec{U})$ and $(B, \vec{V})$ such that: $A(1, 1, \alpha)$, $B(0, 2, 0)$, $\vec{U}(0, 0, \alpha)$ and $\vec{V}(\beta, 3, 0)$ where α and β are real numbers. Let the torsor $T = (A, \vec{U}) + (B, \vec{V})$.

1. Give the reduction elements of T at point O .
2. What is the necessary and sufficient condition for T to be a slider?
3. Determine the support of T .

Correction

Exercise 6:

Let three sliders $(A_i, \vec{V}_i)$; $i = 1, 2, 3$ such that: $A_1(0, 0, a)$, $A_2(a, 0, 0)$ and $A_3(0, a, 0)$; $\vec{V}_1(2a, 0, 0)$, $\vec{V}_2(0, 3a, 0)$ and $\vec{V}_3(0, 0, \lambda a)$. We consider the torsor $L = \sum_{i=1}^3 (A_i, \vec{V}_i)$. Determine the reduction elements of L at an arbitrary point P as well as its central axis.

Correction

Exercise 7:

Consider, in a Cartesian frame $R(O, X, Y, Z)$ with basis $(\vec{i}, \vec{j}, \vec{k})$, the following system of three bound vectors defined by:

$$\begin{aligned} A_1(1, 0, 0) \quad \vec{V}_1 &= a\vec{i} + b\vec{j} + \vec{k} \\ A_2(0, 1, 0) \quad \vec{V}_2 &= 2b\vec{i} - 2a\vec{j} - 3\vec{k} \\ A_3(0, 0, 1) \quad \vec{V}_3 &= -8\vec{i} + \vec{j} + c\vec{k} \end{aligned}$$

where a , b , and c are real constants.

For what values of a , b , and c is the system of three bound vectors equivalent to a couple? Determine then its moment.

Correction

Exercise 8:

We consider the vector field $\vec{W}$ which associates to every point $M(x, y, z)$ the vector $\vec{W}(M)$ with components:

$$W_x = -1 - y - 2az$$

$$W_y = 3 + x - 2az$$

$$W_z = -2 + 2ax + a^2y$$

where a is a real parameter.

1. For what values of a is the field $\vec{W}$ an antisymmetric field?
2. Determine, in each case, the reduction elements of the associated torsor.

Correction